

Uncertainty Quantification

How the Bryan η -meson R_{CP} thesis estimates, combines, and propagates its uncertainties

A systematic restructuring of Chapter 4 (*Uncertainty Estimations*)
and the R_{CP} propagation in Chapter 5 of
*J. W. Bryan, “Measuring single particle suppression of the η meson
using the sPHENIX detector in Au + Au collisions at $\sqrt{s_{NN}} = 200$ GeV”*
(pre-defense draft, 26 Nov 2025)

This document reorganizes the thesis’s uncertainty treatment into a single, logically ordered pipeline: (i) what the measurement is, (ii) every individual source of uncertainty and how its size is obtained, (iii) how sources are combined inside each ingredient, and (iv) how everything propagates into the final observable, R_{CP} . Equation and table numbers in parentheses refer to the thesis. Nothing here changes the physics; it only re-sequences and annotates it.

1 The big picture: what is being measured

The observable is the nuclear modification factor built as a ratio of two fully corrected invariant yields — central over peripheral:

$$R_{\text{CP}}(p_{\text{T}}) = \frac{\left[\frac{1}{N_{\text{evt}}} \frac{1}{2\pi p_{\text{T}}} \frac{1}{\epsilon} \frac{1}{0.3936} \frac{dN}{dp_{\text{T}}} \right]^{0-20\%} / \langle N_{\text{coll}} \rangle^{0-20\%}}{\left[\frac{1}{N_{\text{evt}}} \frac{1}{2\pi p_{\text{T}}} \frac{1}{\epsilon} \frac{1}{0.3936} \frac{dN}{dp_{\text{T}}} \right]^{60-92\%} / \langle N_{\text{coll}} \rangle^{60-92\%}}. \quad (1)$$

Because R_{CP} is a ratio, the uncertainty bookkeeping has to be done on the *ingredients* that go into each spectrum, then carried through the ratio. The thesis identifies four ingredients that carry uncertainty, plus the global N_{coll} normalization:

Taxonomy of uncertainty sources

Ingredient	Systematic driver	Statistical driver
Reconstruction efficiency ϵ (from MC)	Kinematic-cut variation ($\Delta R, \alpha, \chi^2$) via <i>envelope</i> method	Binomial statistics of found vs. truth η
Energy scale (EM-Cal)	Centrality-dependent $\pm$ shift of the absolute scale	— (treated as pure systematic)
Raw yield in data	Kinematic-cut variation ($\alpha, \chi^2, \Delta R_{\gamma\gamma}$) + photon conversions	$\sqrt{S + 2B}$ statistics background-subtraction
N_{coll} scaling	Glauber/PHENIX N_{coll} relative errors	— (global normalization, not point-by-point)

The pipeline at a glance

1. **Per ingredient, per source:** get an *absolute* error in each fine p_T bin and centrality.
2. **Within an ingredient:** combine its independent sources in *quadrature*.
3. **Rebin:** collapse the fine 0.5 GeV simulation bins into the wider data p_T bins (2–3, 3–4 GeV, ...).
4. **Per spectrum:** combine efficiency, energy-scale, and yield errors in quadrature to a single relative systematic per p_T bin.
5. **Into R_{CP} :** propagate central and peripheral relative errors through the ratio; keep the N_{coll} piece separate as a *global normalization box*.

2 Ingredient 1 — Reconstruction efficiency (Monte Carlo)

The efficiency ϵ is the geometric $\times$ reconstruction efficiency: the fraction of truth η ’s that are successfully reconstructed in simulation. It carries both a systematic and a statistical uncertainty.

2.1 Systematic part: cut variation + the envelope method

The baseline analysis applies a set of kinematic cuts. To probe how sensitive ϵ is to those choices, each cut is independently moved to a *tight* and a *loose* setting while all other cuts are held at baseline. The variations (thesis Table 4.1) are:

Setting	ΔR	α	χ^2	$\Delta R_{\gamma\gamma}$
Tight	0.05	0.6	3	0.9
Baseline	0.075	0.7	4	1.0
Loose	0.10	0.8	5	1.1

For each varied cut the deviation from baseline is taken as an *absolute* error, e.g. for the tight variation of ΔR (thesis Eq. 4.1):

$$\sigma_{\text{tight}} = |\Delta R_{\text{baseline}} - \Delta R_{\text{tight}}|, \quad (2)$$

and analogously for the loose variation.

The envelope method (the key idea)

For each p_T bin, each centrality, and each cut, take the *numerically larger* of the two absolute errors (tight vs. loose). This single “worst-case” deviation per cut is deliberately conservative — it is the half-width of the envelope spanned by the cut variation. The independent cut envelopes are then summed in quadrature to a single efficiency systematic (thesis Eq. 4.2):

$$\sigma^{\text{sys}}(p_T, \text{cent}) = \sqrt{\sigma_{\text{max}, \Delta R}^2 + \sigma_{\text{max}, \alpha}^2 + \sigma_{\text{max}, \chi^2}^2}. \quad (3)$$

2.2 Statistical part: binomial efficiency error

Because ϵ is a fraction of successes (reconstructed η 's) out of trials (truth η 's), its statistical error follows binomial statistics. Starting from the binomial variance of the count X (thesis Eq. 4.3),

$$\text{Var}(X) = N p (1 - p), \quad (4)$$

the efficiency is the scaled variable $\bar{X} \equiv X/N$, so (thesis Eq. 4.4)

$$\text{Var}\left(\frac{X}{N}\right) = \frac{1}{N^2} N p (1 - p), \quad (5)$$

which in terms of the measured efficiency ϵ and truth count N gives the variance (thesis Eq. 4.5) and its square root, the per-bin statistical error (thesis Eq. 4.6):

$$\text{Var}(\epsilon) = \frac{\epsilon(1 - \epsilon)}{N}, \quad \sigma^{\text{stat}}(p_{\text{T}}, \text{cent}) = \sqrt{\frac{\epsilon(1 - \epsilon)}{N}}. \quad (6)$$

2.3 Rebinning the fine simulation bins into data bins

Fine $\rightarrow$ coarse binning

Simulation efficiency is computed in fine 0.5 GeV p_{T} steps, but the data analysis uses wider (~ 1 GeV) bins. Each data bin therefore spans two simulation bins (e.g. the 2–3 GeV data bin covers two 0.5 GeV sim bins). The final efficiency *and its errors* for each data bin are obtained by combining the contributing fine bins; the thesis states the systematic error is *averaged* across the fine bins that fall inside each data p_{T} bin.

Annotation: this is the one step where the thesis prose is least explicit. “Combining bin data” for the central value of an efficiency is most defensibly done by summing matched and generated counts across the sub-bins and re-taking the ratio (so the binomial error in Eq. 4.6 follows naturally), rather than by averaging two ratios. If you are reproducing this for π^0 , state which convention you use.

Two figures are shown per centrality (thesis Figs. 4.1–4.3): the *left* panel plots ϵ in the native 0.5 GeV steps with absolute systematic (red band) and statistical (vertical bars) errors; the *right* panel plots the same after rebinning to the data p_{T} bins, now expressed as *relative* systematic (red) and statistical (blue) errors.

3 Ingredient 2 — EMCal energy scale

The energy-scale uncertainty captures how accurately the EMCal converts measured deposits into an absolute photon energy (calibration constants, detector response, non-linearity). The π^0 is the primary absolute-scale calibrant, though η can also be used.

How the energy-scale error is assigned

1. A $p+p$ prompt-photon study established a 2.6% scale uncertainty. Because this analysis uses Au + Au data with an *initial* calibration, 2.6% was deemed too small, so more conservative, centrality-dependent shifts were adopted:

$$\pm 4\% \text{ (central)}, \quad \pm 3.5\% \text{ (minimum bias)}, \quad \pm 3\% \text{ (peripheral)}.$$

2. Each shift is applied to the spectrum, and the shifted spectrum is divided by the *baseline fit* (a modified Hagedorn function) to read off the relative error vs. p_T (thesis Fig. 4.4).
3. Because the $+$ and $-$ shifts give an *asymmetric* band, the final error is the magnitude of the average of the two one-sided bands (thesis Fig. 4.5).

Resulting relative energy-scale errors: central ≈ 18 –29%, peripheral ≈ 14 –23%, minimum bias ≈ 16 –26%. This is treated as a pure systematic (no statistical counterpart).

4 Ingredient 3 — Raw yield in data

4.1 Systematic part: cut variation (no ΔR) + envelope

The same tight/loose cut philosophy and envelope method used for the MC efficiency are applied to the data yields, with one difference: ΔR (proximity of truth photons to reconstructed clusters) is not available in data, so the data systematic uses α , χ^2 , and $\Delta R_{\gamma\gamma}$. The envelope (largest absolute deviation) and quadrature combination are identical in spirit to Eqs. (4.1)–(4.2).

4.2 Statistical part: background-subtraction error

Yields follow Poisson statistics, but the mixed-event background is scaled in the side-bands of the invariant-mass distribution, and *each* side-band contributes an error. The resulting statistical error on the extracted signal yield is (thesis Eq. 4.7)

$$\sigma_S = \sqrt{S + 2B}, \quad (7)$$

where S is the signal and B the background under the peak — the “ $+2B$ ” accounting for the two-sided background subtraction.

4.3 Reading the dominance plots

For each centrality the thesis plots a *relative fractional* uncertainty: each source’s value divided by the sum of all sources, showing which systematic dominates vs. p_T (thesis Figs. 4.6–4.8). Qualitative findings:

- **Central (0–20%):** χ^2 or α dominate; $\Delta R_{\gamma\gamma}$ and signal contribute little. Cut errors span 10–70%; statistical < 1 –41%.
- **Peripheral (60–92%):** no single cut dominates across p_T ; the signal error dominates the last bin. Higher S/B keeps systematics smaller (few % to $\sim 40\%$); statistical < 1 –77%.
- **Minimum bias (0–92%):** between the two; α usually the largest cut. Up to $\sim 60\%$ at low p_T , falling to $\sim 20\%$ at mid p_T .

4.4 Photon-conversion uncertainties

Photons can convert to e^+e^- pairs. Both internal and external conversions are carried in the analysis chain, each contributing a yield uncertainty:

- **Internal** ($\gamma \rightarrow e^+e^-$, branching ratio 0.7%): a conservative flat 1% uncertainty applied across all p_T .
- **External:** no correction applied; instead a flat 3% yield uncertainty, based on a previous sPHENIX study.

5 Ingredient 4 — N_{coll} global normalization

This analysis adopts PHENIX N_{coll} values (from the former BBC). Crucially these are *global normalization* uncertainties, *not* point-by-point errors — they shift all data points of a centrality class together.

Centrality [%]	0–5	5–10	10–15	15–20	60–65	65–70	70–75	75–80	80–85	85–90
N_{coll}	1062	850.1	675.1	533.5	35.0	23.4	15.5	9.9	6.2	3.9
Rel. err [%]	6.8	6.4	6.2	5.9	6.5	7.3	7.2	5.4	6.0	6.1

Table 1: PHENIX N_{coll} and relative errors over the sub-bins (thesis Table 3.3; 90–95% sub-bin omitted for width).

The 0–20% and 60–92% N_{coll} values are the arithmetic means over their sub-bins; the relative errors are quadrature-summed. The resulting global normalization uncertainties are **12.67%** (central) and **17.49%** (peripheral), drawn as a box on the R_{CP} result that represents how far *all* points may shift coherently.

6 Combining into final spectra uncertainties

For each centrality, the efficiency, energy-scale, and yield systematics are combined in quadrature to a single relative systematic per p_{T} bin, and the statistical pieces likewise. Systematics dominate across nearly all centralities and p_{T} bins (driven by the kinematic-cut variations); the lone exception is the highest peripheral bin, where statistics blow up. The thesis reports:

- **Central:** systematic $\sim 30\%$ at low/mid p_{T} rising to $\sim 73\%$; statistical $\sim 3\%$ up to $\sim 10\%$, with the last bin at 41% (small data volume).
- **Peripheral:** systematic 10–50% (no extreme outliers); statistical 2–9%, jumping to 19% (10–12 GeV) and 77% (last bin).
- **Minimum bias:** systematic $\sim 30\%$ (up to 60% in the 2–3 and 4–5 GeV bins); statistical $\leq 6\%$ until the last bin (20%).

The full per-centrality bookkeeping is tabulated in the thesis (Tables 4.2–4.4). The 0–20% table is reproduced below as a representative example.

p_{T} [GeV]	Raw Yield	Raw stat [%]	Raw sys [%]	ϵ	ϵ stat [%]	ϵ sys [%]	E_{scale} [%]	Inv. stat [%]	Inv. sys [%]
2.5	1.09×10^6	0.10	20.43	0.07	4.53	21.45	20.87	4.53	29.66
3.5	3.07×10^5	0.18	37.50	0.11	3.44	22.75	25.52	3.44	43.96
4.5	6.13×10^4	0.40	23.65	0.13	3.11	24.55	28.94	3.14	34.31
5.5	3.35×10^4	0.55	9.67	0.15	2.87	27.23	31.43	2.92	29.28
6.5	1.08×10^4	0.96	17.38	0.16	2.66	27.43	33.20	2.83	32.93
7.5	4.27×10^3	1.53	26.46	0.17	2.65	30.10	34.40	3.06	40.49
8.5	1.88×10^3	2.31	30.42	0.17	2.63	28.49	35.14	3.50	42.09
9.5	5.51×10^2	4.26	50.80	0.18	2.49	30.07	35.51	4.94	59.39
11	2.24×10^2	9.45	67.55	0.21	1.62	28.96	35.51	9.58	73.87
14	2.37×10^1	41.07	61.84	0.24	1.04	26.37	34.05	41.08	67.73

Table 2: Invariant yields and relative uncertainties for 0–20% centrality (thesis Table 4.2). The final invariant-yield systematic (last column) is the quadrature combination of the raw-yield, efficiency, and energy-scale systematics.

7 Propagation into the final observable R_{CP}

The fully corrected invariant yield (thesis Eq. 5.5) applies, per centrality c ,

$$\frac{1}{N_{\text{evt}}} E \frac{d^3 N}{dp^3} = \frac{1}{4\pi p_T} \frac{1}{N_{\text{evt}}^c} \frac{1}{\epsilon^c} \frac{1}{0.3936} \frac{dN_{\eta}^c}{dp_T}, \quad (8)$$

where $1/0.3936$ is the $\eta \rightarrow \gamma\gamma$ branching-ratio correction (PDG). R_{CP} is then the ratio of central to peripheral corrected spectra. The point-by-point systematic and statistical errors of the two spectra are propagated through the ratio; the N_{coll} uncertainty is *not* folded in point-by-point but shown as a single global box.

Final R_{CP} uncertainties (thesis Table 5.1)

p_T [GeV]	2–3	3–4	4–5	5–6	6–7	7–8	8–9	9–10	10–12	12–16
Sys [%]	43.36	47.80	41.84	38.65	43.73	56.11	63.00	79.94	83.48	84.21
Stat [%]	4.94	3.68	3.37	3.27	3.66	4.74	6.76	10.10	21.50	87.42

Systematics dominate and grow with p_T (43–84%); statistical errors stay low (3.3–21%) except the last 12–16 GeV bin (87%). The orange box at unity on the R_{CP} plot is the peripheral global-normalization uncertainty from PHENIX’s N_{coll} (the conservative choice of the larger, peripheral value), interpreted as a coherent shift of all points.

Why systematics rise at mid-to-high p_T

The thesis attributes the larger systematics around 5–9 GeV partly to background mis-estimation as π^0 *merging* sets in: different polynomial degrees used to model the background give larger variance there, and that fit-model spread feeds directly into the cut-variation systematic.

8 One-page summary: the full recipe

End-to-end uncertainty recipe

1. **Efficiency systematic.** Vary $\Delta R, \alpha, \chi^2$ tight/loose; per bin take the larger absolute deviation (envelope); quadrature-sum the three: $\sigma^{\text{sys}} = \sqrt{\sigma_{\text{max}, \Delta R}^2 + \sigma_{\text{max}, \alpha}^2 + \sigma_{\text{max}, \chi^2}^2}$.
2. **Efficiency statistical.** Binomial: $\sigma^{\text{stat}} = \sqrt{\epsilon(1 - \epsilon)/N}$.
3. **Rebin** fine 0.5 GeV sim bins into data p_T bins (average the systematic across the fine bins inside each data bin).
4. **Energy scale.** Apply centrality-dependent $\pm$ shifts (4/3.5/3%); ratio shifted spectrum to baseline (Hagedorn) fit; take magnitude of the average of the two asymmetric bands.

5. **Yield systematic.** Same envelope method on $\alpha, \chi^2, \Delta R_{\gamma\gamma}$ (no ΔR in data).
6. **Yield statistical.** $\sigma_S = \sqrt{S + 2B}$ (two-sided side-band subtraction).
7. **Conversions.** +1% (internal) and +3% (external) flat yield systematics.
8. **Per spectrum** quadrature-combine efficiency $\oplus$ energy-scale $\oplus$ yield to one relative systematic and one statistical per p_T bin (Tables 4.2–4.4).
9. N_{coll} . Quadrature-sum sub-bin relative errors $\rightarrow$ 12.67% (central), 17.49% (peripheral); keep as a *global normalization box*, not point-by-point.
10. R_{CP} . Propagate central/peripheral relative errors through the ratio (Table 5.1); draw the peripheral N_{coll} box at unity.

Prepared as a structured reference to Bryan's uncertainty methodology — useful as an anchor for an analogous π^0 R_{CP} systematics treatment.